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PHASE II – NATIONAL LEVEL TECHNICAL FEST
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PRESTIGE IS ABOVE ALL



Presented by

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Team Profile

TRACK / PROBLEM STATEMENT

TERRA02 - Industrial Pollution Source Attribution Engine

TEAM NAME

DeveloperX

TEAM LEADER'S NAME

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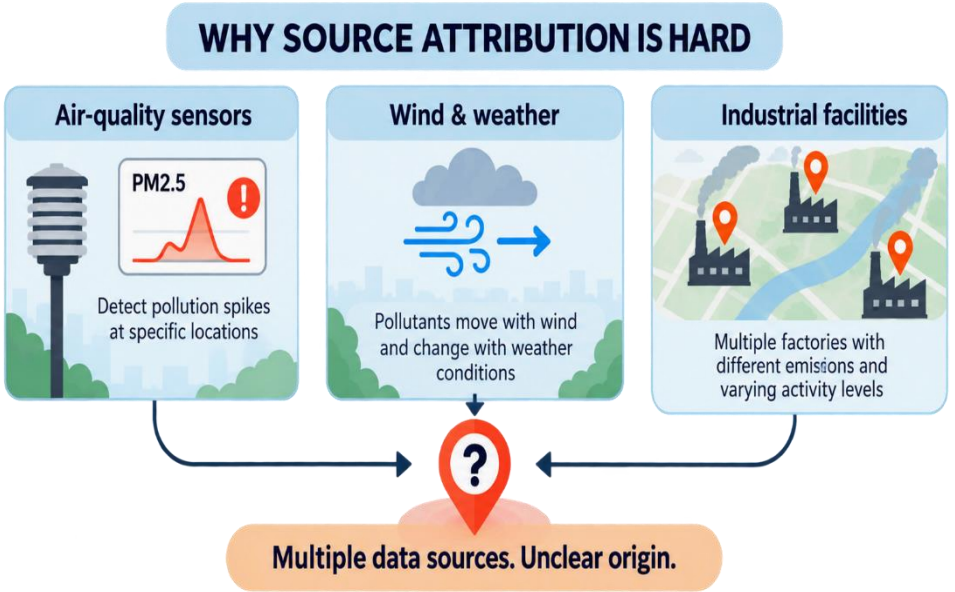
TEAM MEMBERS

Afsar Azam – Model Integration & Frontend & Backend
Lalita Jhapate – Preprocessing & Model Training
Ayush Singh - Deployment & Documentation

Problem Statement - Air-quality sensors detect pollution, but identifying the industrial sources responsible is difficult because pollution movement depends on wind, weather, location and time.

Pollution Detected. Source Unknown

- **Unidentified Sources:** Sensors detect pollution but cannot pinpoint the responsible factory.
- **Scattered Data:** Air quality, wind, and factory data are disconnected.
- **Slow Investigation:** Finding likely sources requires time and analysis.
- **Uncertain Evidence:** Lack of clear evidence makes attribution difficult



Key Challenge: Which factories are the likely sources of pollution, and what evidence supports it?

Solution Approach – AIR KAVACH - From Pollution Detection to Explainable Source Attribution

- **COLLECT**

Gather air-quality, weather, wind, and factory-location data.

- **ANALYZE**

Detect pollution spikes and analyze wind patterns.

- **IDENTIFY**

Screen nearby factories using distance and wind alignment.

- **PRIORITIZE**

Rank likely sources and assess evidence reliability.

- **EXPLAIN & REPORT**

Present source rankings, evidence, and uncertainty.

Detect the spike. Trace the source. Explain the evidence.



Workflow



AIR KAVACH transforms environmental observations into structured investigation insights by combining air-quality data, weather analysis, geospatial screening and evidence assessment to identify potential industrial pollution sources.

Collect → Preprocess → Detect → Analyse → Screen → Validate → Investigate

Key Features & Innovation

KEY FEATURES

01. Multi-Source Integration
Combines air-quality, weather, wind, and facility data.

02. Pollution Detection
Identifies unusual pollution spikes.

03. Wind & Geospatial Analysis
Analyzes wind patterns and facility distances.

04. Source Ranking
Ranks potential industrial sources using available evidence.

05. Evidence & Uncertainty
Highlights evidence gaps and data limitations.

06. Automated Reporting
Generates structured investigation reports.



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Technology Stack

01 FRONTEND, BACKEND & DATA
- React.js
- Vite
- Python
- FastAPI
- Pandas
- NumPy

02 AI / ML & ANALYTICS
- Scikit-learn
- Isolation Forest — Anomaly Detection
- Pollution Event Analysis
- Wind Analysis
- Geospatial Source Screening

03 APIS & INTEGRATIONS
- OpenAQ
- Open-Meteo
- CPCB / Data.gov.in
- FastAPI REST APIs

04 CLOUD / DEPLOYMENT
- Docker

SYSTEM ARCHITECTURE
AIR KAVACH

React + Vite
Frontend

↓

FastAPI Backend
API, Business Logic & Data Handling

↓

Environmental Data Processing
Air Quality, Weather, Wind & Facility Data (Pandas + NumPy)

↓

ML & Geospatial Screening
Anomaly Detection, Wind Analysis & Source Screening

↓

Candidate Source Ranking
Rank Potential Industrial Sources using Available Evidence

↓

Investigation Workspace & Reports
Interactive Analysis, Evidence View & Report Generation

Feasibility & Viability

FEASIBILITY

- TECHNICAL FEASIBILITY

Uses Python, FastAPI, Machine Learning and geospatial analysis to investigate potential pollution sources.

- DATA AVAILABILITY

Can use available air-quality observations, weather data and industrial facility locations.

- IMPLEMENTATION FEASIBILITY

Supports pollution event detection, wind analysis, candidate source screening and evidence report generation.

- SCALABILITY

Designed to support different regions when compatible environmental data and facility locations are available.

VIABILITY

- COST-EFFECTIVE DEVELOPMENT

Uses open-source technologies such as Python, FastAPI and Scikit-learn to reduce software development costs.

- ACCESSIBLE DATA SOURCES

Can leverage available public environmental datasets, subject to access, licensing and API limitations.

- PRACTICAL APPLICATION

Helps environmental analysts prioritize potential pollution sources and organize investigation evidence.

- FUTURE EXPANSION

Can be extended to additional regions as reliable air-quality, weather and industrial facility data become available.

Impact & Benefits

IMPACT

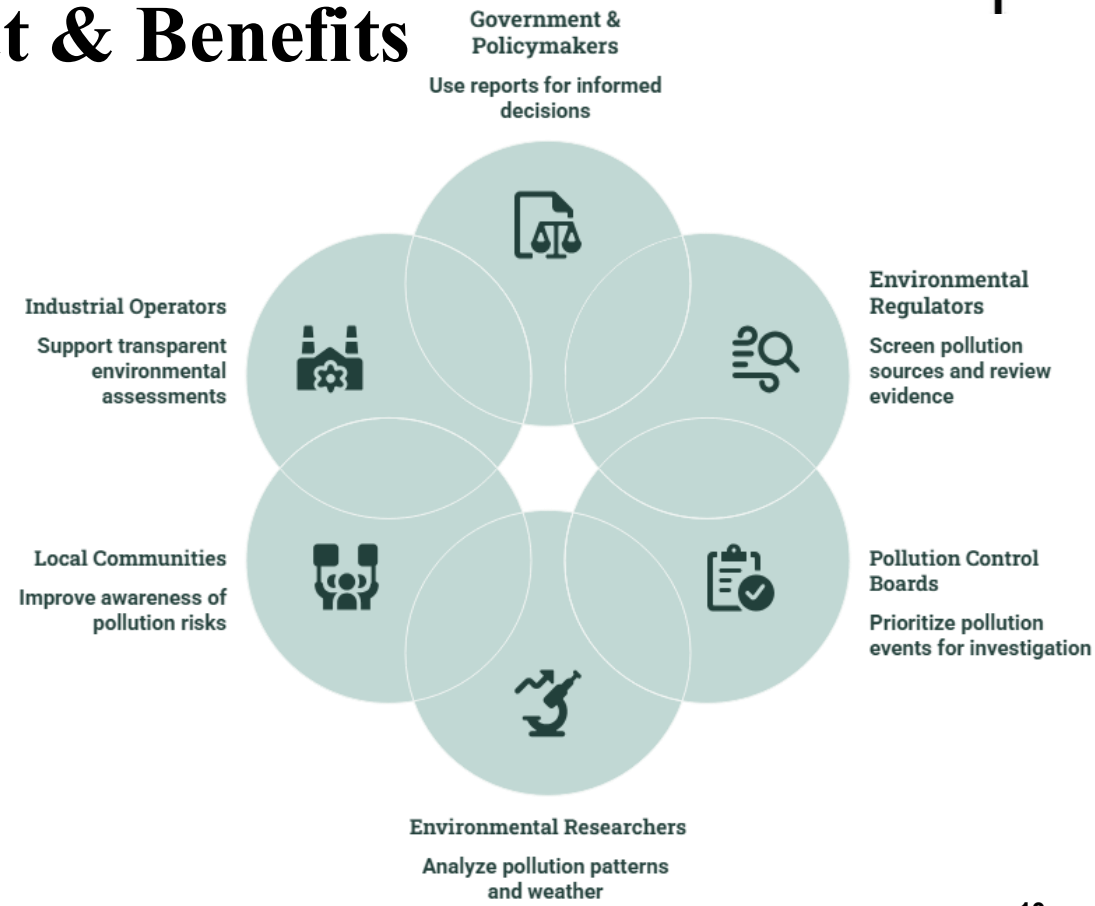
01. ENVIRONMENTAL PROTECTION
Supports industrial pollution investigations.

02. INFORMED ENVIRONMENTAL ACTION
Helps prioritize pollution investigations.

03. PUBLIC HEALTH AWARENESS
Improves awareness of pollution risks.

04. RESPONSIBLE INDUSTRIAL MONITORING
Promotes evidence-based source assessment.

05. REGIONAL ENVIRONMENTAL INTELLIGENCE
Supports expansion across different regions.



References

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Official API Documentation

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Climate Change Service (C3S).

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Central Pollution Control Board

OpenStreetMap Contributors
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Leaflet Documentation

Scikit-learn Documentation

GeoPandas Documentation

FastAPI Documentation

